

The First Assignment:

Group Details:

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Project Title (**Maximum 10 words**):

AI-Based Performance Analysis and Prediction Model for Al-Jamoom High Schools

Project Description (**Minimum 100 words**):

This project uses practical AI techniques to look into high school data from AL-Jamoom Governorate and predict future academic performance. Education quality matters a lot to the local community, so having an early view of how schools are doing helps administrators act quickly when issues show up.

We gathered real numbers from past years, mainly looking at student averages in Qudrat and Tahsili exams, along with official school evaluation results. By running simple machine learning model like Linear Regression and Random Forests, our system picks up trends from older data to predict upcoming test performance. This gives school managers in AL-Jamoom a clear idea of which schools need extra support or better resources.

Define the AI technologies that will be applied in your project? (**Minimum 500 words**):

- 1- Data Preprocessing and Normalization : This is the preliminary stage that precedes feeding data into the algorithms used to build artificial intelligence; it consists of two phases:

The first is preprocessing : transforming raw data—which is typically messy, incomplete, unformatted (numbers written as words), and unprepared in real-world scenarios—into a format that the machine can understand and process.

The second step is normalization: a mathematical technique applied to the data organized and cleaned in the first step. It aims to standardize the numerical scale for instance, compressing values to a range between 0 and 1 to ensure unbiased comparisons between different features. For example, if comparing an age feature (0 to 100) against an "income" feature (measured in the thousands), or an academic score (out of 100) against a school level (1 to 31), normalization ensures that algorithms assign equal importance to every feature in the project, preventing the model from becoming biased toward features with larger numerical values.

When we began analyzing high school data in Al-Jumum Governorate, we quickly realized that simply gathering numbers was insufficient for generating accurate predictions. To build a reliable system that helps local administrators and decision-makers predict academic performance, we had to subject this raw data to a crucial preparatory phase known as Data Preprocessing and Normalization.

Consequently, we collected official records from all boys' and girls' high schools in the Al-Jumum Governorate sector; while the data was difficult to obtain... We obtained results from the General Achievement Test and the Aptitude Test spanning the last three years, alongside official school assessment data. However, we encountered common data-related challenges regarding clarity and organization. For instance, some schools lacked data for specific years, or there were human errors in data entry. This is where the normalization algorithm comes into play a process that transforms raw, scattered data into a format ready for analysis. We carefully addressed missing values to fill gaps and excluded anomalies and outliers that could mislead models and lead to erroneous conclusions about a school's performance level.

Once the data was cleaned and verified for completeness, another challenge emerged: the significant disparity in numerical scales. Aptitude and Achievement test scores are typically based on a scale of 100, whereas official school assessments often rely on much smaller scales. If we were to feed these raw figures directly into an algorithm like Linear Regression, the model would automatically prioritize the Achievement scores simply because the numbers are larger, thereby ignoring the profound impact of the school assessment.

To resolve this issue, we applied a "normalization" step. Through this mathematical process, we compressed and standardized the ranges of all inputs to a uniform, equitable scale. This step is crucial to ensure that machine learning algorithms—whether Linear Regression or Random Forests—evaluate the impact of each factor on equal footing, assigning fair weight to school assessments just as they do to standardized test scores.

Ultimately, the effort invested in cleaning and normalizing the data for Al-Jumum Governorate... It is the solid foundation upon which we built the accuracy of our projections. This effort transformed scattered historical data into a clear, reliable picture, providing the Education Department with early, precise insight into which schools are on the right track and which require urgent intervention and additional resource allocation to improve the quality of education.

2- Multiple Linear Regression : Multiple Linear Regression is a straightforward machine learning method that predicts a single number (like an upcoming test score) by looking at several factors at once. Instead of relying on just one clue, it combines multiple inputs and figures out how much each one actually matters.

The formula is basically an expanded line equation:

$$Y_{\text{prime}} = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$$

How we apply it to our Al-Jamoom project:

Instead of guessing blindly, we use this model to predict how high schools in Al-Jamoom will perform on

upcoming standardized tests based on their past track record:

What we want to predict (Y):

The school's expected average score in the upcoming Tahsili exam.

The clues we feed into the model (X):

X1: The school's past average in the Qudrat exam.

X2: The school's past average in the Tahsili exam.

X3: The school's official rating from the Education and Training Evaluation Commission (ETEC).

X4: The average internal high school GPA.

What the weights (w) tell us:

The weights are the most practical part for school administrators. They show the real-world impact of each factor on its own. For instance, the weight for ETEC (w_3) tells us:

"If a school improves its operational rating by one level, how many points will that add to its students' test scores?"

Why we start with this model:

We picked Multiple Linear Regression as our starting baseline because it isn't a "black box." It gives local supervisors in Al-Jamoom a clear, readable formula that shows whether a school is falling behind because of basic student performance or because the school environment itself needs immediate support.

3- Random Forest : We selected Random Forest because relying on a single decision tree usually leads to predictive mistakes. If a model uses only one tree, it easily overfits the training data. Random Forest resolves this issue by building an ensemble of many trees, where each tree analyzes a random subset of the dataset. At the end of the process, all trees vote, and the model selects the most common outcome.

For our Al-Jamoom school project, this algorithm provides support in two practical ways:

School Categorization: It groups high schools into distinct categories, clearly showing which institutions perform well and which require extra administrative attention.

Feature Importance: It calculates the exact factors that drive student results in Qiyas exams. This allows us to determine whether academic success is most affected by teacher availability, leadership quality, or school resources.

Seeing these specific driving factors makes it significantly easier for local educational authorities in Al-Jamoom to make data-driven decisions and allocate help exactly where it should go first.

References:

1. García, S., Luengo, J., & Herrera, F. (2015). Data Preprocessing in Data Mining. Springer.
2. Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied Linear Statistical Models*. McGraw-Hill.
3. <https://youtu.be/gkXX4h3qYm4>

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